# Lyra Self-Audit Cal #35 on Composite v0.3 Substrate-Deficit Term N_c$^{4/2}$N_c

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~13:50 EDT

## Self-Audit Cal #35 on Composite v0.3 Substrate-Deficit Term

### 0. Goal

Per Lyra Composite v0.3 substantive substrate-clean candidate (~13:42 EDT) + Cal #35 STANDING (Independence-Taxonomy-Before-Multiplicative-Null-Model + Cal Calibration #35 form-selection ≠ mechanism-forcing) + Cal #36 STANDING positive-search rule + Cal #27 STANDING CLAIMS-tier discipline: substantive Cal-style self-audit on Composite v0.3 substrate-deficit term N_c^4 / 2^N_c substrate-natural form.

### 1. Composite v0.3 Substantive Candidate

Per Lyra Composite v0.3:

$$n_C \cdot (n_C/rank)_3 + N_c^4/2^{N_c} = 1575/8 + 81/8 = 1656/8 = 207 \text{ substrate-natural}$$

vs T190 = $(24/\pi^2)$^{C_2} ≈ 206.755 → composite v0.3 substrate-natural form at 0.112% precision to observed m_$\mu$/m_e ≈ 206.768.

### 2. Term-by-Term Substrate-Mechanism Audit

Term 1: n_C · (n_C/rank)_3

| Property | Status |
|---|---|
| substrate-natural form | n_C ratio scaling + Pochhammer at substrate-Bergman kernel exponent n_C/rank substrate-natural |
| Substrate-mechanism FORCING derivation | substrate-Bergman matrix element at Sym^3 V_fund substrate-K-type V_(3/2, 1/2) per Schur II Pochhammer rule + substrate-Bergman kernel substrate-natural exponent n_C/rank = 5/2 |
| Independent substrate-mechanism | YES — substrate-Bergman + substrate-Pochhammer Schur II independently substantive derivation per FK Ch. XII §VI + Helgason Ch. IX multi-week joint Lyra+Keeper+Elie |
| Cal #35 STANDING audit | substantive substrate-mechanism FORCING candidate at Tier 2 STRUCTURAL precision per substrate-Bergman + substrate-Pochhammer derivation |

Status: Term 1 substantive substrate-mechanism FORCING candidate; multi-week explicit derivation per Cal #189.

Term 2: N_c^4 / 2^N_c

| Property | Status |
|---|---|
| substrate-natural form | substrate-color N_c substrate-natural to 4th power + substrate-Cartan 2^N_c substrate-natural denominator |
| Substrate-mechanism FORCING derivation | NO independent substrate-mechanism derivation visible v0.3 (substantive observation only) |
| Independent substrate-mechanism | UNCLEAR — N_c^4 substrate-natural form requires substrate-mechanism for 4th power substrate-natural exponent at gen-2 V_(3/2, 1/2) |
| Cal #35 STANDING audit | substantive form-fitting POSITIVE SEARCH candidate per Cal #36 STANDING NOT substrate-mechanism FORCING form |

Status: Term 2 substantive form-fitting candidate per Cal #35 STANDING; substantive substrate-mechanism FORCING form requires substantive substrate-mechanism for N_c^4 substrate-natural exponent at gen-2 V_(3/2, 1/2) per Casey #13 per-generation substrate-mechanism class multi-week per Cal #189.

## 3. Composite Substantive Cal #35 Honest Verdict v0.1

Per Cal #35 STANDING audit: - Term 1 (n_C · (5/2)_3 = 196.875): substantive substrate-mechanism FORCING candidate at Tier 2 STRUCTURAL via substrate-Bergman + substrate-Pochhammer Schur II - Term 2 (N_c^4 / 2^N_c = 10.125): substantive form-fitting candidate; substrate-mechanism FORCING form for N_c^4 substrate-natural exponent UNRESOLVED v0.3

Cal #35 STANDING honest verdict: Composite v0.3 substrate-natural form 207 substrate-natural at 0.112% precision is POSITIVE SEARCH candidate per Cal #36 STANDING for Term 1 + form-fitting candidate per Cal #35 STANDING for Term 2.

Substrate-mechanism FORCING form: substantive PARTIAL — Term 1 substrate-Bergman + substrate-Pochhammer multi-week candidate; Term 2 substrate-deficit per Casey #13 per-generation substrate-mechanism class multi-week candidate.

Substantive precision 0.112%: Tier 1 / Tier 2 boundary substrate-natural; substantive precision floor per Two-Tier substrate-precision hypothesis (Elie Toy 3648 Saturday May 24) at structural mass/mixing observables.

## 4. Substantive Substrate-Mechanism Multi-Week Path

Per Cal #35 STANDING honest verdict v0.1:

Term 1 substrate-mechanism multi-week: - substrate-Bergman normalization Z_K-type per substrate-K-type V_(3/2, 1/2) explicit derivation per FK Ch. XII §VI joint Lyra+Keeper+Elie - substrate-Pochhammer Schur II (n_C/rank)_k = (5/2)*3 substrate-natural form per Sym^k k=3 substrate-K-type V*(3/2, 1/2) substantive derivation

Term 2 substrate-mechanism multi-week: - substrate-deficit per Casey #13 per-generation substrate-mechanism class substantive substrate-natural form N_c^4 substrate-natural exponent at gen-2 V_(3/2, 1/2) - substrate-Cartan 2^N_c substrate-natural denominator substantive substrate-mechanism per substrate-K-type substrate-natural form

Substantive composite multi-week per Cal #189: substrate-Bergman + substrate-Pochhammer + substrate-deficit composite substantive substrate-mechanism FORCING form derivation explicit.

## 5. Closure v0.1

Self-Audit Cal #35 on Composite v0.3 substrate-deficit term N_c^4 / 2^N_c v0.1:

Substantive honest verdict per Cal #35 STANDING: Composite v0.3 substrate-natural form 207 at 0.112% precision is PARTIAL substrate-mechanism FORCING candidate: - Term 1 (n_C · (5/2)_3): substrate-mechanism FORCING candidate via substrate-Bergman + substrate-Pochhammer Schur II multi-week - Term 2 (N_c^4 / 2^N_c): substantive form-fitting candidate; substrate-mechanism FORCING form per Casey #13 per-generation substrate-mechanism class multi-week

Substantive precision 0.112%: Tier 1 / Tier 2 boundary substrate-natural per Two-Tier substrate-precision hypothesis (Elie Toy 3648).

Audit-chain Lyra-on-self brake event 6 Thursday afternoon: self-audit on substantive substrate-clean composite v0.3 candidate per Cal #35 STANDING + Cal #36 STANDING + Cal #27 STANDING discipline → substantive PARTIAL substrate-mechanism FORCING candidate honest verdict v0.1.

Per Cal #35 STANDING + Cal #36 STANDING + Cal #27 STANDING + Cal #189: substantive 6th audit-chain Lyra-on-self brake event Thursday afternoon — self-audit on substantive substrate-clean candidate v0.3 per Cal #35 STANDING independence-taxonomy-before-multiplicative-null-model operational standing rule.

Tier: Composite v0.3 PARTIAL substrate-mechanism FORCING candidate honest verdict per Cal #35 STANDING + Cal #36 STANDING; substantive multi-week explicit substrate-mechanism derivation per Cal #189.

— Lyra, Thu 2026-06-04 ~13:50 EDT. Self-Audit Cal #35 on Composite v0.3 v0.1: Term 1 (n_C · (5/2)_3) substantive substrate-mechanism FORCING candidate via substrate-Bergman + substrate-Pochhammer Schur II; Term 2 (N_c^4 / 2^N_c) substantive form-fitting candidate per Cal #35 STANDING; substantive composite v0.3 PARTIAL substrate-mechanism FORCING candidate; substantive precision 0.112% Tier 1/Tier 2 boundary per Two-Tier hypothesis Elie Toy 3648; substantive 6th audit-chain Lyra-on-self brake event Thursday afternoon.